Fundamentals of STL

Thursday, March 12, 2026 10:20 PM

C++ → in Built $\boxed{\text{STL}}$

① Pair

```
pair<int, string> demoPair;
```

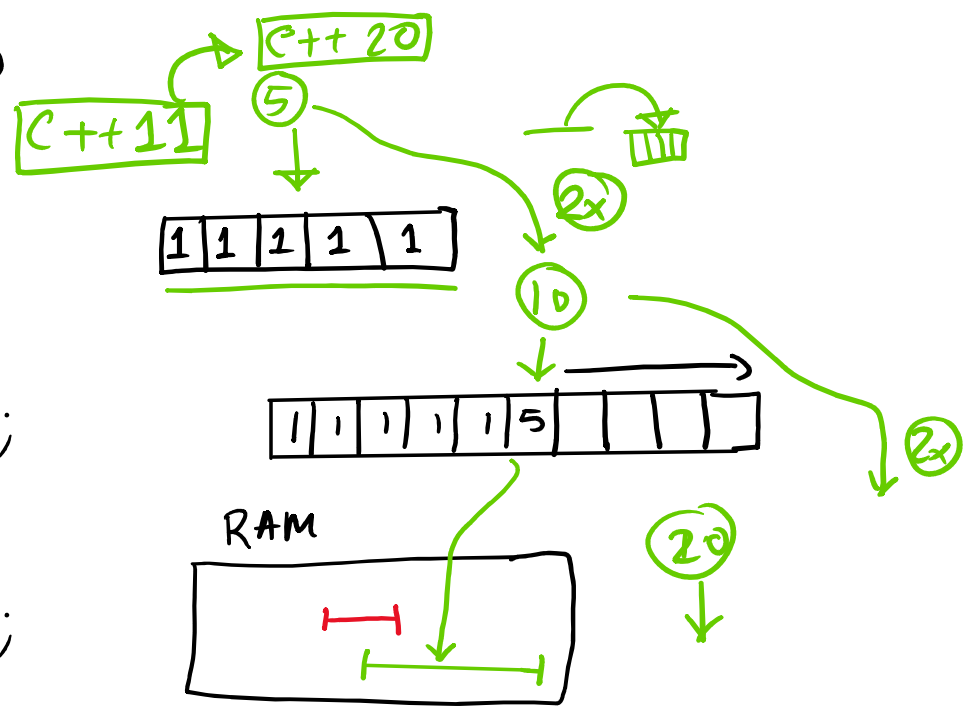
{1, "Karim"}

② Vector

```
int arr[10];
vector<int> vec;
vec.push_back(1)
vec.pop_back()
vec.size() → 0
```

[ ]

```
vector<int> vec;
vec.push_back(1);
vec.push_back(1);
vec.push_back(1);
vec.push_back(1);
vec.push_back(1);
vec.push_back(5);
```



$\approx \boxed{O(1)}$

③ Unordered_Set / HashSet

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[1, 2, 3]

{1, 2, 3, 2} → {1, 3, 2}

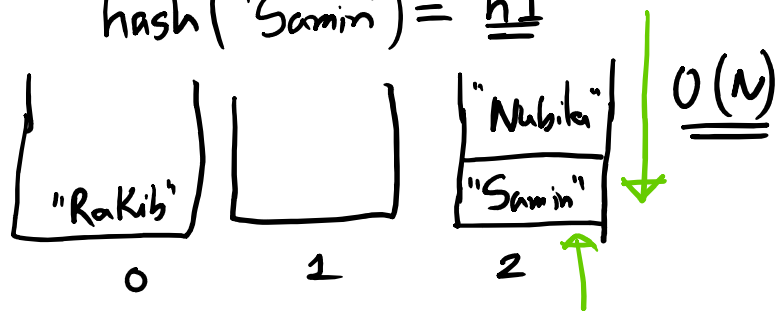
Hash Bucket

["Samin", "RaKib", "Nabila"]

hash("Nabila") = h1

hash("RaKib") = h2

hash("Samin") = h1



Avg: O(1)

worstCase → O(N)

O(N + N) ≈ O(N)

h1 % 3 = 2

h2 % 3 = 0

insert() → Avg. = O(1) worse Case = O(N)

erase() →

count() → O(M)

O(N·M)

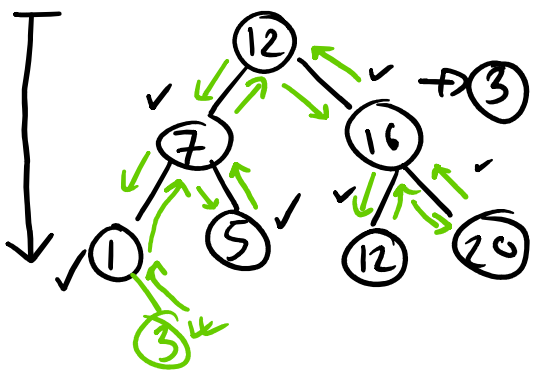
size() → O(1)

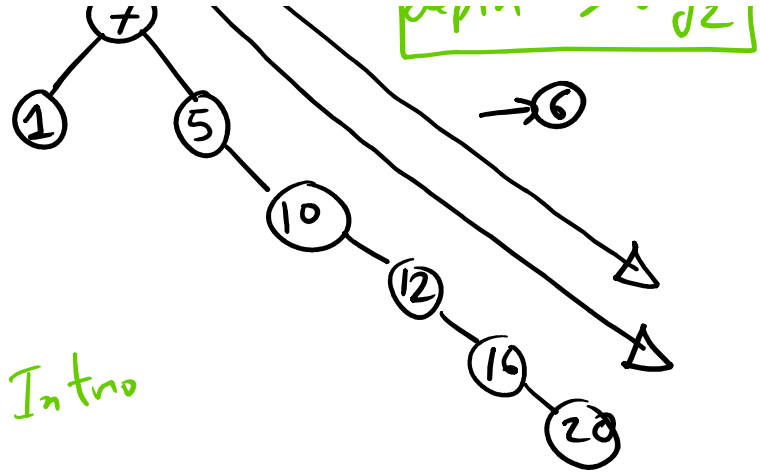
④ Set

Self Balancing Binary Search Tree / Red-Black Tree

pre order

depth > $\log_2 N$





1, 5, 7, 12, 16, 10 sort( ) → Intro

$\boxed{\log_2(N)}$

insert( ) → $O(\log N)$

erase( ) → $O(\log N)$

count( ) → $O(\log)$

size( ) → $O(1)$

⑥ Unordered Map/Hash Map

Generics → Java

vector<T> vec;

double, int, string

Int, double, String

S<T> value

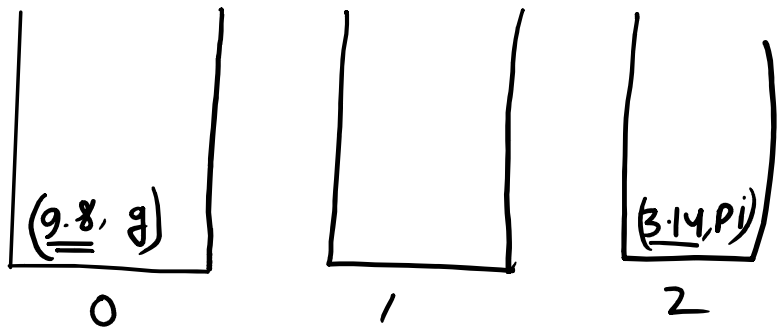
(Key, Value)

insert → O(1) / O(N)

erase

find

count



hash(Key) = H1

H1 % 3